1. Tick those options that fall within the definition of a 'biosignature'. 1 point

- [x] Evidence for environmental conditions that may support life
- [ ] Evidence for current biological activities
- [ ] Signs of past life

2. How has life influenced the composition of the terrestrial atmosphere? 1 point

- ( ) It has contributed greatly to the photolysis of carbon dioxide
- (•) Biological activities have steered it away from chemical equilibrium
- ( ) It has been driven towards a state of equilibrium rather than away from it

3. Which of the following are used as biosignatures in exoplanet research? Tick all that apply. 1 point

- [ ] Water vapor
- [ ] Surface temperature
- [x] Certain atmospheric gases
- [x] Evidence for vegetation

4. Which of these could represent biosignatures of photosynthesizing plants on a planet's surface? 1 point

- [ ] Nitrous oxide
- [x] Oxygen or ozone
- [x] A 'red edge' signal
- [ ] Liquid water

5. Select those biosignature gases that would be most applicable to detecting signs of life in an extraterrestrial environment resembling the Early Earth (3.5 - 4 billions of years ago). 1 point

- [x] Methane
- [x] Ethane
- [ ] Carbon dioxide
- [ ] Oxygen

6. Which of the following sentences about spectroscopy is true? 1 point

- ( ) Land-based telescopes are incapable of measuring atmospheric spectra
- ( ) It is entirely based on measurements of visible light
- (•) It measures how radiation is absorbed and/or transmitted across a range of wavelengths
- ( ) Spectra can be used to infer the presence but not the relative concentrations of different atmospheric gases

7. If one discovered organic compounds within the atmosphere of an exoplanet, what could be concluded? 1 point

- [x] Depending on the compounds, they could function as 'building blocks' for carbon-based life
- [x] Some of the compounds could represent atmospheric biosignature gases
- [ ] The planet is habitable, even if there is no direct evidence for life
- [ ] Life must be present in one form or another

8. Why are combinations of biosignature gases in a non-equilibrium state useful to life detection missions? 1 point

- (•) They constitute more convincing evidence for biology than individual biosignatures
- ( ) Gases in a non-equilibrium state are uninfluenced by atmospheric conditions
- ( ) Only gases which are near their chemical equilibrium are useful to life detection
- ( ) They are particularly easy to detect by spectroscopy

9. How is biogenic methane formed? 1 point

- ( ) It is a by-product of photochemistry
- (•) Microorganisms produce it by breaking down organic matter
- ( ) It is produced by cyanobacteria when there is little oxygen available

10. What is the name of the instrument used to obtain the first evidence for organic compounds in an extrasolar atmosphere? 1 point

- ( ) James Webb Space Telescope
- ( ) James Clerk Maxwell Space Telescope
- (•) Hubble Space Telescope
- ( ) Evidence for organics has only been found within our Solar System

11. Of the following types of radiation, which is likely to be the most useful to detecting signs of life on an exoplanet? 1 point

- ( ) Gamma radiation
- ( ) X-ray radiation
- ( ) None of the above
- (•) Infrared radiation

12. Which of the following statements is false? 1 point

- (•) Detecting biosignatures is the primary aim of most current exoplanet missions
- ( ) Oxygen can be produced non-biologically
- ( ) Ethane is considered a potential biosignature gas

13. When did we first discover Earth-sized exoplanets orbiting a star similar to our Sun? 1 point

- (•) In 2011
- ( ) In 2011, but only one such planet has been identified
- ( ) Approximately a decade ago
- ( ) Planets like these are yet to be discovered

14. What does it mean when certain molecules are referred to as 'prebiotic'? 1 point

- (•) They precede the development of life
- ( ) They contain living organisms (e.g. bacteria)
- ( ) Life has contributed to their formation over millions of years

15. Name one of the main contributions of the space telescope Kepler to our search for extraterrestrial life. 1 point

- ( ) It was the first space telescope to characterize the atmosphere of an extrasolar planet
- ( ) It was the first telescope to identify water vapour in the atmosphere of an exoplanet
- (•) It has enabled the discovery of more than fifty rocky exoplanets

16. Which of the following should be considered when selecting potential biosignatures to measure? 1 point

- [x] The planetary surface temperature
- [x] The presence of carbon dioxide
- [x] Availability of liquid water

17. Why is the infrared signal of ozone used as a proxy for the presence of atmospheric oxygen? 1 point

- (•) Ozone is detectable at infrared wavelengths
- ( ) Oxygen cannot be detected by spectroscopy
- ( ) Concentrations of oxygen are usually too low to measure directly

18. Name one of the advantages of studying exoplanets that partially eclipse their parent star (as seen from the Earth). 1 point

- (•) It provides a way to separate atmospheric spectra from the background signal
- ( ) The analysis of biosignatures is only possible for these types of planets
- ( ) These types of planets are easy to study because there is less background noise

19. Which of the following statements about land-based telescopes is true? 1 point

- (•) They may enable direct analysis of Earth-like exoplanets in the future
- ( ) They are the only way to study exoplanet atmospheres
- ( ) They can only be used to study very large gas giant exoplanets

20. What is expected from exoplanet missions in the future? Tick all that apply. 1 point

- [x] Tens of thousands of new planetary systems may be discovered
- [x] There will be an increased focus on life detection
- [x] Both land- and space-based telescopes will be used